

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE PHASE-III

TEST # 01

DATE : 15-07-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 01	Topics	DATE : 15 - 07 - 2018
PRE-MEDICAL NURTURE COURSE : PHASE - III		
PHYSICS : Basic Maths, Vector		
CHEMISTRY : Mole Concept		
BIOLOGY : Animal Tissues, Cell Biology		

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. $\cos(120^\circ)$ equals to:

- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) $\frac{\sqrt{3}}{2}$ (4) $-\frac{\sqrt{3}}{2}$

2. $\sin\left(\frac{11\pi}{6}\right)$ equal to :

- (1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{2}$ (3) $-\frac{1}{2}$ (4) $-\frac{\sqrt{3}}{2}$

3. Value of $(\sin^2 30^\circ - 1) + (\cos^2 30^\circ - 1) = \dots$

- (1) 1 (2) -1 (3) 2 (4) -2

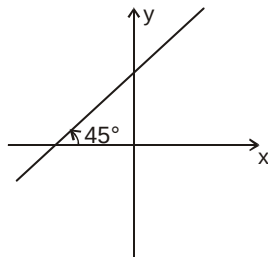
4. $\sin 22.5^\circ \cdot \cos 22.5^\circ$ equal to:

- (1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{8}}$

5. 1 radian is equal to -

- (1) $\frac{\pi}{180}$ degree (2) $\frac{180}{\pi}$ degree
(3) $\frac{90}{\pi}$ degree (4) $\frac{18}{\pi}$ degree

6. Slope of line shown in the figure is:



- (1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{\sqrt{3}}{2}$

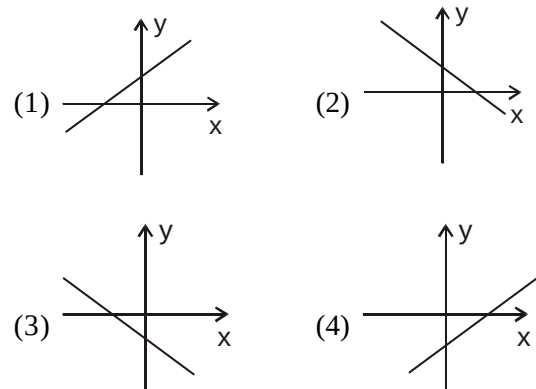
7. Of $y = x + \frac{1}{x}$, then $\frac{dy}{dx}$ is

- (1) $1 + \frac{1}{x^2}$ (2) $x - \frac{1}{x^2}$
(3) $1 - \frac{1}{x^2}$ (4) $x^2 - \frac{1}{x^2}$

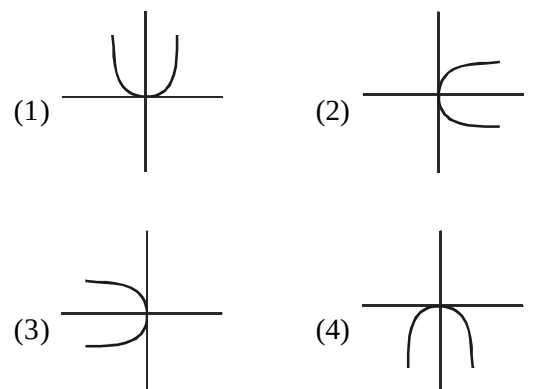
8. If $y = x \ln x$, then $\frac{dy}{dx}$ is

- (1) $\ln x - 1$ (2) $\ln x + x$
(3) $\ln x + 1$ (4) $x \ln x + 1$

9. Correct graph for $y = 5x + 3$ is



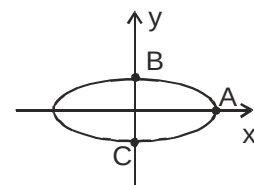
10. Correct graph for $y^2 = 4ax$, where $a = \text{constant}$ may be:



11. $(0.97)^{\frac{1}{3}}$ is approximately equals to:

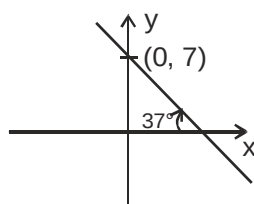
- (1) 0.99 (2) 0.01
(3) 0.985 (4) None

12. For ellipse $\frac{x^2}{36} + \frac{y^2}{25} = 1$, the co-ordinates of points A and C are respectively:



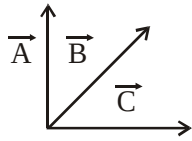
- (1) (6, 0) and (0, 5) (2) (36, 0) and (25, 0)
(3) (6, 0) and (0, -5) (4) (25, 0) and (0, 36)

13. If $y = \log_e e^{3x}$, then $\frac{d^2y}{dx^2}$ is
 (1) 0 (2) 1
 (3) e^x (4) $\frac{1}{e^x}$
14. Condition for maximum value of a function:
 (1) $\frac{dy}{dx} \neq 0, \frac{d^2y}{dx^2} < 0$ (2) $\frac{dy}{dx} \neq 0, \frac{d^2y}{dx^2} > 0$
 (3) $\frac{dy}{dx} = 0, \frac{d^2y}{dx^2} > 0$ (4) $\frac{dy}{dx} = 0, \frac{d^2y}{dx^2} < 0$
15. If $y = \log(ax + b)$ then $\frac{dy}{dx}$ is
 (1) $\frac{a+b}{ax+b}$ (2) $\frac{a}{ax+b}$
 (3) $\frac{b}{ax+b}$ (4) $(ax + b)a$
16. If $y = 3\sin^2x + 3\cos^2x$, then $\frac{dy}{dx}$ is:
 (1) $6\sin x + 6\cos x$ (2) $3\sin^2x$
 (3) $6\sin x - 6\cos x$ (4) zero
17. $\frac{d}{dx} \sin(\log x)$
 (1) $\frac{\cos x}{x}$ (2) $\cos(\log x)$
 (3) $\frac{\cos(\log x)}{x}$ (4) None
18. The minimum value of $y = 4x^2 - 16x + 9$ is:
 (1) 7 (2) 0
 (3) 2 (4) -7
19. In which quadrant " $\sin \theta$ " is negative :-
 (1) II (2) IV
 (3) III (4) Both II & III
20. For given graph of straight line. Which statement is correct:



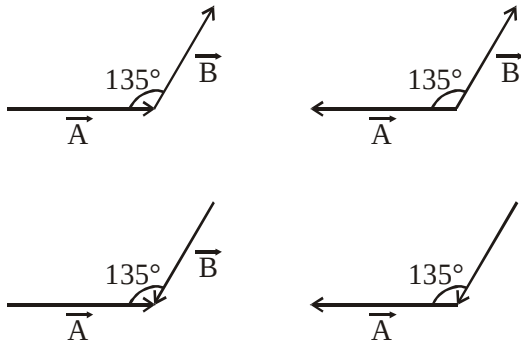
- (1) Slope angle is 143°
 (2) Intercept on y-axis is 7
 (3) Slope is negative
 (4) all above
21. $\sin(5^\circ)$ equal to:
 (1) $\frac{\pi}{180}$ (2) $\frac{\pi}{90}$ (3) $\frac{\pi}{36}$ (4) None
22. Evaluate the integral $\int_1^2 \frac{1}{r^3} dr$:
 (1) $-\frac{1}{2}$ (2) $\frac{1}{2}$ (3) $-\frac{3}{8}$ (4) $\frac{3}{8}$
23. If $y = \sin x + e^{2x}$ then find value of $\left(\frac{dy}{dx}\right)_{at x=0}$:-
 (1) 0 (2) 2
 (3) $\cos x + 2e^{2x}$ (4) 3
24. If $y = \frac{x^2}{(x+1)}$ then $\frac{dy}{dx}$ is :
 (1) $\frac{x(x+2)}{(x+1)^2}$ (2) $\frac{x(x+2)}{(x+1)}$
 (3) $\frac{x^2 - 2x}{(x+1)^2}$ (4) $\frac{x^2 - 1}{(x+1)^2}$
25. If $S = t^2 + 2t - 1$ the find $\left(\frac{dS}{dt}\right)_{at t=1}$?
 (1) 2 (2) 4 (3) 0 (4) 1
26. If $y = \sin x - \cos x$ then $\frac{d^2y}{dx^2}$ is :
 (1) $\cos x - \sin x$ (2) $\cos x + \sin x$
 (3) $\sin x - \cos x$ (4) $-\cos x - \sin x$
27. $\frac{d}{dx}(\log_e x^2) = \dots\dots\dots$
 (1) $\frac{1}{x}$ (2) $\frac{2}{x}$ (3) x^2 (4) e^x
28. $\frac{1}{10} \frac{d}{dx}(e^{5x}) = \dots\dots\dots$
 (1) $\frac{1}{2} e^{5x}$ (2) $\frac{1}{5} e^{5x}$ (3) $\frac{1}{15} e^{5x}$ (4) $\frac{x}{5}$

29. $\vec{A}, \vec{B}$ and $\vec{C}$ are :-



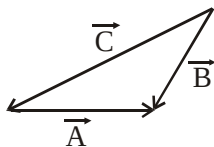
- (1) Concurrent vectors
(2) co-initial vectors
(3) Both (1) and (2)
(4) Parallel vectors

30. What is the angle between following vectors :-



- (1) $45^\circ, 135^\circ, 135^\circ, 45^\circ$
(2) $135^\circ, 135^\circ, 45^\circ, 45^\circ$
(3) $135^\circ, 45^\circ, 45^\circ, 135^\circ$
(4) $135^\circ, 45^\circ, 45^\circ, 45^\circ$

31. Identify correct statement :-



- (1) $\vec{A} + \vec{B} = \vec{C}$ (2) $\vec{B} + \vec{C} = \vec{A}$
(3) $\vec{C} + \vec{A} = \vec{B}$ (4) $\vec{A} + \vec{B} + \vec{C} = 0$

32. The resultant of 2 forces acting at an angle of 120° is 10N and is perpendicular to one of the force. The smaller force is :-

- (1) $\frac{10}{3}$ (2) $\frac{20}{3}$ (3) $\frac{10}{\sqrt{3}}$ (4) $\frac{20}{\sqrt{3}}$ N

33. If two forces act in opposite direction their resultant is 1N and if they act mutually perpendicular their resultant is 5N find magnitude of both forces :-

- (1) 8N, 7N (2) 5N, 4N
(3) 5N, 6N (4) 3N, 4N

34. The angle between the two vectors $-2\hat{i} + 3\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$ is :-

- (1) 0° (2) 90°
(3) 180° (4) None of these

35. The vector $\vec{OA}$ where O is origin is given by $\vec{OA} = 2\hat{i} + 2\hat{j}$. Now it is rotated by 45° anti-clockwise about O. What will be the new vector :-

- (1) $2\sqrt{2}\hat{j}$ (2) $2\hat{j}$
(3) $2\hat{i}$ (4) $2\sqrt{2}\hat{i}$

36. Minimum number of unequal coplanar forces whose vector sum can be equal to zero is :-

- (1) Two (2) Three (3) Four (4) Any

37. Out of the following the resultant of which cannot be 4 newton ?

- (1) 2N and 2N (2) 1N and 3N
(3) 2N and 6N (4) 2N and 8N

38. If $\vec{A} = 5\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - c\hat{k}$ are perpendicular vectors, the value of c is :-

- (1) +2 (2) 4
(3) -7 (4) +8

39. The angle made by the vector $4\hat{i} - 3\hat{j} + 5\hat{k}$ with z-axis is :-

- (1) 30° (2) 45° (3) 90° (4) 120°

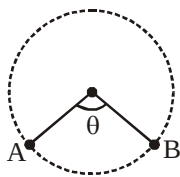
40. The vector component of the vector $\vec{A} = 2\hat{i} + 3\hat{j}$ on vector $\vec{B} = \hat{i} + \hat{j}$ is :-

- (1) 5 (2) $\frac{5}{\sqrt{2}}$ (3) $5\sqrt{2}$ (4) 10

41. $\frac{d}{dx}(a^x)$ equals.

- (1) xa^{n-1} (2) $\frac{a^x}{\ln a}$
(3) $a^x \ln a$ (4) none of these

42. A particle is moving on a circular path with a constant speed 'v'. Its magnitude of change in velocity as it moves from A to B is :-



- (1) $2v \sin \theta$ (2) $v \sin \theta/2$
(3) $v \cos \theta$ (4) $2v \sin \theta/2$
43. $0.8\hat{i} + 0.6\hat{j} + c\hat{k}$ represents a unit vector when c is :-
(1) 0 (2) $\frac{2\sqrt{3}}{5}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{2}{5\sqrt{3}}$
44. Find value of $10_{10^4} 16^2$. Given that $(\log_{10} = 0.3)$
(1) 0.6 (2) 0.9 (3) 1.2 (4) 1.5
45. $\frac{d}{dx} \log_{10} 10 =$
(1) $\frac{1}{10}$ (2) 100 (3) 1 (4) zero
46. The total number of protons in 2 g of CH_4 is :-
(1) 10 (2) 6.02×10^{24}
(3) 7.525×10^{23} (4) 12×10^{23}
47. How many moles of electrons weight one kilogram?
(1) 6.023×10^{23} (2) $\frac{1}{9.108} \times 10^{31}$
(3) $\frac{6.023}{9.108} \times 10^{54}$ (4) $\frac{1}{9.108 \times 6.023} \times 10^8$
48. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of urea solution is :-
(1) 0.001 M (2) 0.1 M
(3) 0.02 M (4) 0.01 M
49. Which of the following has the largest number of molecules?
(1) 1 mol N_2
(2) 22.4 L CO_2 at STP
(3) 180 g glucose
(4) All are having equal number of molecules

50. A compound has 28.57% sulphur in it. The minimum molecular mass of the compound is :-
(1) 50 (2) 90 (3) 112 (4) 224
51. 5 mol of A and 12 mol of B are taken for a reaction shown by the equation, $\text{A} + 3\text{B} \rightarrow 2\text{C}$. How many moles of C will be produced?
(1) 5 (2) 12 (3) 24 (4) 8
52. Calculate the molarity of 20 g NaOH in 250 cm^3 aqueous solution :-
(1) 0.2 M (2) 2 M
(3) 0.02 M (4) 20 M
53. In the given Reaction :-
 $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightarrow 2\text{NH}_{3(g)}$
If 3 mole N_2 and 12 mole H_2 is taken then find L.R :-
(1) N_2 (2) H_2
(3) Both (4) None of these
54. Find out average atomic mass of a sample containing ^{10}B and ^{11}B , if percentage of ^{10}B in a mixture is 19%.
(1) 10 (2) 11 (3) 10.81 (4) 10.30
55. Correct relation between vapour density and molar mass is :-
(1) vapour density = $2 \times$ molar mass
(2) molar mass = $2 \times$ vapour density
(3) Both are correct
(4) None of these
56. The maximum quantity of BaSO_4 (white ppt.) produced by mixing 0.5 mol of BaCl_2 and 1 litre of 1 M H_2SO_4 will be :-
(1) 0.5 mol (2) 1.0 mol
(3) 1.2 mol (4) 1.5 mol
57. A mixture containing 200 gm H_2 and 160 gm O_2 is ignited so that water is formed according to the reaction, $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. Which substance is the limiting reagent ?
(1) H_2 (2) O_2
(3) H_2O (4) None of these
58. What is the weight of one atom of H in gram?
(1) 1 g (2) 1.67×10^{-24} g
(3) 2 amu (4) 2 g

59. A gas has molecular formula $(CH)_n$. If vapour density of the gas is 13, what should be the molecular formula of the compound?
 (1) C_3H_3 (2) C_4H_4
 (3) C_2H_2 (4) C_6H_6
60. How many molecules are present in 5.4 gm of glucose ($C_6H_{12}O_6$)?
 (1) 2.05×10^{23} (2) 3.09×10^{22}
 (3) 1.75×10^{23} (4) 1.8×10^{22}
61. What is the weight of 3.01×10^{23} molecules of NH_3 ?
 (1) 8.50 g (2) 17 g
 (3) 34 g (4) 14 g
62. How many carbon atoms are present in 0.35 mole of ($C_6H_{12}O_6$)?
 (1) 1.26×10^{24} (2) 2.1×10^{23}
 (3) 3.3×10^{22} (4) 6.02×10^{22}
63. In a compound X is 70% and Y is 30% by weight percent. If atomic weight of X and Y are 24 and 15 respectively. Then calculate the E.F. of the compound.
 (1) XY_2 (2) X_2Y_5 (3) X_2Y_3 (4) X_3Y_2
64. In V ml of the vapours of substance at NTP weights W g. Then mol. wt. of substance is :-
 (1) $(W/V) \times 22400$ (2) $\frac{V}{W} \times 22.4$
 (3) $(W - V) \times 22400$ (4) $\frac{W \times 1}{V \times 22400}$
65. Molarity equation of a mixture of solutions of same substance is given by :-
 (1) $(M_1 + V_2) \times (M_2 + V_2) \times (M_3 + V_3) + \dots = M_1 + M_2 + M_3$
 (2) $M_1V_1 + M_2V_2 + M_3V_3 + \dots = M(V_1 + V_2 + V_3 + \dots)$
 (3) $\frac{M_1}{V_1} + \frac{M_2}{V_2} + \frac{M_3}{V_3} + \dots = M \left(\frac{1}{V_1} + \frac{1}{V_2} + \frac{1}{V_3} \right)$
 (4) $\frac{M_1}{V_1} \times \frac{M_2}{V_2} \times \frac{M_3}{V_3} + \dots = M_1 \left(\frac{1}{V_1} \times \frac{1}{V_2} \times \frac{1}{V_3} \right)$
66. The group having isoelectronic species is :-
 (1) $O^\ominus, F^\ominus, Na^+, Mg^{2+}$ (2) $O^{2-}, F^-, Na^+, Mg^{2+}$
 (3) O^-, F^-, Na, Mg^+ (4) O^{2-}, F^-, Na, Mg^{2+}
67. Read the following statements :-
 I : Number of atoms present in 18 gm H_2O and in 44 gm CO_2 are equal.
 II : Molecular formula and Empirical formula of a given compound cannot be same.
 III : Number of mole = $\frac{\text{Volume of gas in mL(at STP)}}{22.4}$
 IV : Number of mole = $\frac{N}{N_A}$ (N = number of species)
 Choose the correct option :-
 (1) Statement-I is incorrect
 (2) Statement-IV is incorrect
 (3) Statement-II, III both are incorrect
 (4) All statements are incorrect
68. $5A + 7B + 9C \longrightarrow$ product. If initially in a mixture 15 mole of A, 14 mole of B and 27 mole of C were taken. Which will act as a limiting reagent?
 (1) A (2) B
 (3) C (4) All of these
69. Total number of Neutrons present in 80 gm of D_2O :-
 (1) $60 N_A$ (2) $40 N_A$
 (3) $80 N_A$ (4) $42 N_A$
70. Calculate the mass of 2×10^{22} atoms of Nitrogen?
 (1) 0.56 g (2) 0.47 g
 (3) 1.2 g (4) 1.6 g
71. What will be the mass of 50 atoms of carbon?
 (1) 600 amu (2) 600 gm
 (3) 12 gm (4) 12 amu
72. Two separate sample of O_2 and CH_4 contain their equal weights. What is the ratio of their number of molecules?
 (1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 4 : 7
73. Which one of the following statements is correct for $^{14}_7N$ and $^{15}_7N$?
 (1) Both are having same no. of neutrons
 (2) Both are having same no. of protons
 (3) Both A & B are correct
 (4) none of these

74. The number of electron present in 28 gm of Nitrogen atom (At. wt. of N = 14)
(1) $4N_A$ (2) $14N_A$ (3) $21N_A$ (4) $6N_A$
75. The simplest formula of a compound containing 50% of element X(atomic mass = 10) and 50% of the element Y(atomic mass = 20) by weight is :-
(1) XY (2) X_2Y (3) XY_2 (4) X_2X_3
76. Atomic mass and gram atomic mass of Nitrogen (${}_7N^{14}$) respectively is :-
(1) 14 gm, 14 gm (2) 14 amu, 14
(3) 14 amu, 14 gm (4) 7 amu, 7 gm
77. Number of gm-atoms of sulphur present in SO_2 gas which occupy 22.4 ml at STP.
(1) 0.001 (2) 0.01 (3) 1 (4) 10
78. With increase in temperature which of these changes?
(1) Molarity
(2) Weight fraction of solute
(3) Molality
(4) Mole fraction
79. The total number of atoms of oxygen in 0.2 mol of $Na_2B_4O_7 \cdot 10H_2O$ will be :-
(1) 6.02×10^{23} (2) 1.02×10^{24}
(3) 2.05×10^{24} (4) 2.05×10^{20}
80. For an element, if atomic number (z) and mass number (A) are 15 and 31 respectively. Then the number of protons, neutrons and electrons respectively are :-
(1) 15, 15, 15 (2) 16, 15, 16
(3) 15, 16, 15 (4) 15, 16, 16
81. Four flasks of 1 litre capacity each are separately filled with gases N_2 , Ne, H_2 and CO_2 . At the same temperature and pressure the ratio of the number of atoms of these gases present in different flasks would be :-
(1) 1 : 1 : 1 : 1 (2) 2 : 1 : 2 : 3
(3) 1 : 2 : 1 : 3 (4) 3 : 2 : 2 : 1
82. The number of CO_2 molecule is maximum in :-
(1) 44 gram of CO_2
(2) 44 moles of CO_2
(3) 44 molecules of CO_2
(4) 4.4 gram-molecules of CO_2
83. An oxide of metal M has 40% by mass of oxygen. Metal M has atomic mass of 24. The empirical formula of the oxide.
(1) M_2O (2) M_2O_3 (3) MO (4) M_3O_4
84. The number of mole present in 2 litre of 1.5 M KOH solution is :-
(1) 2 (2) 3 (3) 0.3 (4) 1
85. 1 amu is equal to :-
(1) $\frac{1}{12}$ of C-12 (2) $\frac{1}{14}$ of O-16
(3) 1 gm of H_2 (4) 1.66×10^{-23} kg
86. 0.44 gm of a colourless oxide of nitrogen occupies 224 mL at STP. The compound is :-
(1) N_2O (2) NO (3) N_2O_2 (4) NO_2
87. 1000 gm aqueous solution of $CaCO_3$ contains 10 gm of calcium carbonate. Concentration of solution is :-
(1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10000 ppm
88. 1 mole of CH_4 contains :-
(1) 6.02×10^{23} atoms of H
(2) 4 gm atom of H
(3) 1.81×10^{23} molecular of CH_4
(4) 3.0 gm of carbon
89. $20\% \left(\frac{w}{w} \right)$ aqueous NaOH solution means :-
(1) 20 gm NaOH is dissolved in 100 gm of solvent.
(2) 20 gm NaOH is dissolved in 100 ml of solution
(3) 20 gm NaOH is dissolved in 100 gm of solution
(4) 20 ml NaOH is dissolved in 100 gm solution
90. How much volume of $O_{2(g)}$ is required to completely burn 6 litre of $C_2H_{2(g)}$:-
(1) 10 litre (2) 15 litre
(3) 20 litre (4) 25 litre
91. The term cell was first used by
(1) Robert Brown (2) Robert Hooke
(3) Fontana (4) Walson

92. In 1831, Robert Brown described
 (1) Nucleus (2) Cell
 (3) Nucleolus (4) Nucleoid
93. Who proposed the cell theory ?
 (1) Schleiden and Schwann
 (2) Robert Hooke
 (3) Rudolf Virchow
 (4) Robert Brown
94. The smallest cell of $0.3 \mu\text{m}$ in length is
 (1) Egg of an ostrich
 (2) Nerve cells
 (3) Human RBC
 (4) Mycoplasma
95. Which of the following is incorrectly matched w.r.t. size ?
 (1) Mycoplasma – $0.3 \mu\text{m}$ in length
 (2) Bacteria – $3.5 \mu\text{m}$ length
 (3) RBC – $7 \mu\text{m}$ in diameter
 (4) A typical eukaryotic cell – $1.2 \mu\text{m}$
96. Inclusion bodies
 (1) Reserve food material in eukaryotic cell
 (2) Bounded by any unit membrane
 (3) Are phosphate granules, cyanophycean granules and glycogen granules
 (4) All of these
97. Bacterial flagellum is composed of
 (1) Basal body, axoneme and centrosome
 (2) Basal body, hook and filament
 (3) Axoneme, basal body and hook
 (4) Centrosome, hook and filament
98. The longest portion of bacterial flagellum is
 (1) Pili (2) Filament
 (3) Basal body (4) Hook
99. Bacterial flagella are made up of
 (1) Tubulin protein (2) Flagellin protein
 (3) Pilin protein (4) Nexin protein
100. Prokaryotic cell does not contain
 (1) Mitochondrion (2) Ribosome
 (3) Nucleoid (4) Plasma membrane
101. Mesosomes of prokaryotes perform function similar to
 (1) Peroxisome (2) Mitochondria
 (3) Lysosomes (4) Ribosomes
102. The cell wall of bacteria consist of
 (1) NAG (2) NAM
 (3) Cellulose (4) Both (1) and (2)
103. Single envelope system is characteristic feature of
 (1) Prokaryotic cell (2) Eukaryotic cell
 (3) Mesokaryotic cell (4) All of these
104. Select incorrect match.
 (1) Longest cell – Nerve cell
 (2) Largest isolated single cell – Ostrich egg
 (3) Smallest cell – PPLO
 (4) Spherical bacteria – Bacillus
105. The cell wall of bacteria is composed of
 (1) Cellulose
 (2) Chitin
 (3) Murein
 (4) Pseudopeptidoglycan
106. The main difference in the gram (+) and gram (–) forms of bacteria lies in their
 (1) Cell wall composition
 (2) Nucleoid
 (3) Mesosome formation
 (4) Shape and size
107. Bacteria differ from PPLO in having
 (1) Genophore (2) Cell membrane
 (3) Cell wall (4) Cytoplasm
108. What is the outermost layer of cell envelope in prokaryotic cell ?
 (1) Cell wall (2) Glycocalyx
 (3) Plasma membrane (4) Mesosomes
109. In both Prokaryotic and Eukaryotic cell a semi fluid matrix called.....occupies the volume of cell.
 (1) Cytoplasm (2) Sapvacoule
 (3) Nucleus (4) Glycocalyx
110. Many bacteria have small circular DNA outside the genomic DNA, is called
 (1) Incipient Nucleus
 (2) Prochromosome
 (3) Genophore
 (4) Plasmid
111. Nuclear material without nuclear membrane is observed in
 (1) Bacteria and mycoplasma
 (2) Bacteria and algae
 (3) Cyanobacteria and protist
 (4) Mycoplasma and protozoans

- 112.** Unicellular organisms are made up of single cell and are capable of
 (A) Independent existence
 (B) Performing the essential function of life
 (1) Only (A) is correct
 (2) Only (B) is correct
 (3) Both (A) and (B) are correct
 (4) Neither (A) nor (B) is correct
- 113.** In prokaryotic cell, infoldings of cell membrane is called A which is specialised for B.
 (1) A–Nucleoid, B–Cell division
 (2) A–Mesosomes, B–Cellular respiration
 (3) A–Mesosomes, B–Intracellular digestion
 (4) A–Genophore, B–Cellular respiration
- 114.** Lipopolysaccharide (LPS) Form a layer outside peptidoglycan layer in cell wall in
 (1) Gram (+) bacteria (2) Gram (–) bacteria
 (3) Virus (4) Viroids
- 115.** The infolding of plasma membrane in the form of vesicles in bacterial cell is helpful in
 (1) Cell wall secretion
 (2) DNA duplication
 (3) Increasing surface area for enzymatic activities
 (4) All of these
- 116.** Which of the following controls permeability?
 (1) Cell membrane (2) Cell wall
 (3) Protoplasm (4) Cytoplasm
- 117.** The quasifluid nature of _____ enables _____ movement of proteins within the overall bilayer of a plasma membrane.
 (1) Protein, Lateral
 (2) Protein, Flip-flop
 (3) Lipid, Lateral
 (4) Carbohydrate, Flip-flop
- 118.** The organelles whose major function is storage of hydrolytic enzymes are
 (1) Mitochondria (2) Lysosomes
 (3) Chloroplast (4) Golgi apparatus
- 119.** The major role of Golgi apparatus is
 (1) Glycosidation of lipids
 (2) Glycosylation of protein
 (3) Both (1) and (2)
 (4) Protein synthesis
- 120.** The transfer vesicles from RER fuse with which region of Golgi complex
 (1) Cis (2) Trans
 (3) lateral (4) Maturation face
- 121.** Cell organelle with property of detoxification of drugs is
 (1) RER (2) Lysosome
 (3) Sphaerosome (4) SER
- 122.** Heterophagosome is
 (1) Primary lysosome
 (2) Secondary lysosome
 (3) Residual bodies
 (4) Autophagic Vacuoles
- 123.** The cell recognition and adhesion are facilitated by _____ component of plasma membrane.
 (1) Steroids
 (2) Glycoprotein and glycolipid
 (3) Phospholipids
 (4) Cholesterol
- 124.** Cell plate formation in plant cells is specially performed by
 (1) RER (2) Golgi complex
 (3) SER (4) Lysosome
- 125.** Which of the following cell organelle is a major site for the synthesis of steroidal hormones ?
 (1) Peroxisomes (2) RER
 (3) SER (4) Golgi apparatus
- 126.** The subunits of eukaryotic ribosome are
 (1) 60S + 40S (2) 60S + 30S
 (3) 50S + 30S (4) 50S + 40S
- 127.** Secondary cell wall of plant is
 (1) Located outside the primary wall
 (2) Located just inside middle lamella
 (3) Located inside the primary wall
 (4) Located outside the plasma membrane
- 128.** Which of the following organelles has a continuous connection with outer membrane of nucleus ?
 (1) Golgi apparatus
 (2) Lysosome
 (3) RER
 (4) Mitochondria

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| <p>129. Tonoplast encloses</p> <ol style="list-style-type: none"> (1) Cytoplasm (2) Water, sap, excretory product and other material not useful for the cell (3) Protoplasm (4) Nucleolus <p>130. The fruits after ripening become soft, it is due to the:</p> <ol style="list-style-type: none"> (1) Dissolution of tannin in cell sap (2) Dissolution of middle lamella (3) Formation of ethylene gas (4) Both (1) and (2) <p>131. In plant cells the vacuoles can occupy upto..... percent of the volume of the cell.</p> <p>(1) 30 (2) 90 (3) 45 (4) 40</p> <p>132. Who proposed the fluid mosaic model for plasma membrane ?</p> <ol style="list-style-type: none"> (1) Cramer and Naegeli (2) Robertson (3) Singer and Nicholson (4) Denielli and Davson <p>133. Which of the following is a part of endomembrane system of eukaryotic cell ?</p> <ol style="list-style-type: none"> (1) Endoplasmic reticulum (2) Mitochondria (3) Peroxisomes (4) Chloroplast <p>134. Cis face of Golgi apparatus is</p> <ol style="list-style-type: none"> (1) Concave and forming face (2) Convex and forming face (3) Concave and maturing face (4) Convex and forming face <p>135. In prokaryotes, the genetic material is</p> <ol style="list-style-type: none"> (1) DS circular DNA with Histone protein (2) DS linear DNA without Histone protein (3) DS circular DNA without Histone protein (4) SS circular DNA without Histone protein <p>136. Simple epithelium is made of :-</p> <ol style="list-style-type: none"> (1) Actively dividing cells (2) Several layers of cells (3) Compactly packed single layer of cells (4) Non-cellular layer of hyaluromic acid | <p>137. Goblet cells are :-</p> <ol style="list-style-type: none"> (1) Multicellular glands (2) Stratified epithelium (3) Glial cells (4) Unicellular glands <p>138. Sebaceous glands are :-</p> <ol style="list-style-type: none"> (1) Holocrine (2) Eccrine (3) Merocrine (4) Epocrine <p>139. Inner lining of gut and liver is made of :-</p> <ol style="list-style-type: none"> (1) Simple columnar epithelium (2) Simple cuboidal epithelium (3) Simple squamous epithelium (4) Pseudostratified epithelium <p>140. Stratified and non-keratinized squamous epithelium occurs in :-</p> <ol style="list-style-type: none"> (1) Vagina (2) Buccal cavity (3) Stomach (4) Both (1) and (2) <p>141. Epithelium forming the inner layer of urinary bladder is :-</p> <ol style="list-style-type: none"> (1) Transitional (2) Myoepithelium (3) Neurosensory (4) Simple squamous <p>142. Pseudostratified columnar epithelium is found in :-</p> <ol style="list-style-type: none"> (1) Stomach (2) Vagina (3) Trachea (4) Mouth <p>143. Epithelium present in cornea, oesophagus and vagina is :-</p> <ol style="list-style-type: none"> (1) Stratified columnar (2) Stratified squamous (3) Ciliated (4) Pseudostratified <p>144. Ciliated epithelium lines :-</p> <ol style="list-style-type: none"> (1) Trachea (2) Oesophagus (3) Stomach (4) Duodenum <p>145. When the glandular cells breakdown completely, the glands is known as :-</p> <ol style="list-style-type: none"> (1) Apocrine gland (2) Holocrine gland (3) Eccrine (4) None of the above <p>146. Tessellated epithelium is found in :-</p> <ol style="list-style-type: none"> (1) Lining of blood vessel (2) Lining of intestine (3) Seminiferous tubule (4) Tubular parts of nephron |
|---|--|

147. Germinal epithelium cells are present in :-

- (1) Uriniferous tubules
- (2) Oesophagus
- (3) Inner lining of trachea
- (4) Seminiferous tubules

148. Microvilli of epithelium cells

- (1) Push mucus
- (2) Engulf the foreign matter
- (3) Give movement to the cells
- (4) Increase surface area

149. The ducts of most of the glands and tubular parts of nephron in kidneys are lined by :-

- (1) Cuboidal epithelium
- (2) Columnar epithelium
- (3) Stratified squamous
- (4) Ciliated squamous

150. Which of the following is the main function of compound epithelium?

- (1) Protection
- (2) Absorption
- (3) Reproduction
- (4) Secretion

151. Which statements is correct regarding compound epithelium?

- (1) All the cells rest on the basement membrane
- (2) Single layer of cells
- (3) It consists of the similar type of cells
- (4) It is mainly protective

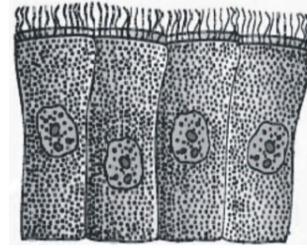
152. Choose the correct location of the lining epithelium which lacks basement membrane

- (1) Lining of small intestine
- (2) Lining of ureter
- (3) Lining of buccal cavity
- (4) Wall of blood vessel

153. The cuboidal epithelium cells of have microvilli which increase absorption surface area :-

- (1) oviduct
- (2) Proximal convoluted tubule
- (3) Gall bladder
- (4) Blood vessel

154. Following is the diagram of an epithelium, Choose the correct option w.r.t. the location where this type of epithelium is found :-



- (1) Bronchioles
- (2) Fallopian tubes
- (3) Ventricles of brain
- (4) All of these

155. Which of the following is **incorrect**?

- (1) Gap junction allow rapid transfer of ions
- (2) Adhering junctions are also called desmosomes
- (3) Tight junctions prevent leakage of material across the tissue
- (4) Gap junctions are also called zonula occludens

156. On the basis of mode of secretion, which of the following glands release their secretion without any damage to intact cell ?

- (1) Mammary glands
- (2) Goblet cells
- (3) Sudorific gland of armpits
- (4) Sebaceous gland

157. Stratified epithelium has many layer of cells, however the deepest layer is made up of columnar or cuboidal cells. Thus epithelium is classified on the basis of :-

- (1) Shape of the cells present in the middle layer
- (2) Total number of layers
- (3) Shape of the cells present in the superficial layer
- (4) Shape of the cells in the lower most layer

158. The term histology, i.e. study, of tissue was given by

- (1) Malpighi
- (2) Mayer
- (3) Harvey
- (4) Bichat

159. Which of the following statement is not applicable for pseudostratified epithelium?

- (1) It appears like multilayered because nuclei lie at different level
- (2) The long cells have oval nuclei however short cells have rounded nuclei
- (3) Short cells have cilia at their free surface, however the long cells are without cilia
- (4) It is found in trachea

160. If we scrape the living of buccal cavity by sterilized forceps, what type of cells are is likely to be observed in the microscope?

- (1) Non-keratinized stratified squamous epithelium
- (2) Straitified columnar epithelium
- (3) Pseudustraitified epithelium
- (4) Keratilized stratified squamous epithelium

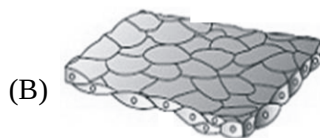
161. Which of the following cell junction perform cementing to keep the neighbouring cells together?

- (1) Gap junction
- (2) Tight junction
- (3) Plasmodesmata
- (4) Adhering junction

162. Apocrine secretion of gland means

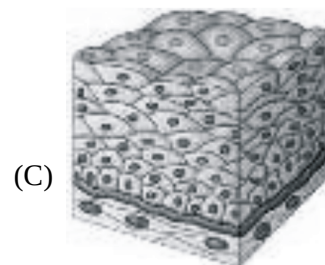
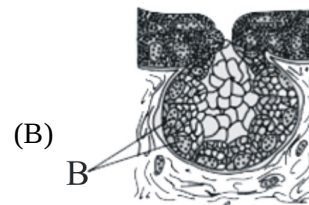
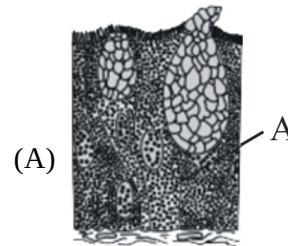
- (1) When part of apical cytoplasm is lost
- (2) When the product is released, the cell remain intact
- (3) Entire contents of cells is discharged with the destruction of cell
- (4) None of these

163. Identify, A, B and C in the diagram given below :-



	A	B	C
(1)	Simple columnar	Simple squamous	Simple ciliated
(2)	Simple squamous	Simple columnar	Simple cuboid
(3)	Simple cuboidal	Simple squamous	Simple ciliated columnar
(4)	Simple cuboidal	Simple columnar	Simple ciliated cuboidal

164. Identify A, B and C in the given diagram and choose the correct combination of options :-



	A	B	C
(1)	Unicellular gland	Multicellular gland	Transitional epithelium
(2)	Multicellular gland	Unicellular gland	Stratified squamous epithelium
(3)	Unicellular gland	Multicellular gland	Stratified squamous epithelium
(4)	Multicellular gland	Unicellular gland	Transitional epithelium

- 165.** The outermost sheath of connective tissue that surrounds a skeletal muscle is :-
 (1) Endomysium
 (2) perimysium
 (3) Epineurium
 (4) Epimysium
- 166.** Perichondrium covers the :-
 (1) Cartilage
 (2) Muscles
 (3) Bone
 (4) Decalcified bone
- 167.** Hyaline cartilage is present at :-
 (1) Tip of the nose
 (2) Epiglottis
 (3) Articular ends of long bones
 (4) The ends of digits
- 168.** Calcified cartilage is present in :-
 (1) Supra scapula of pectoral girdle of frog
 (2) Ends of bones
 (3) Nasal septum
 (4) Pinna
- 169.** The main difference between bone and cartilage is of :-
 (1) Matrix with protein
 (2) Lacunae
 (3) Mineral salts
 (4) Haversian canals
- 170.** Bone has longitudinal canals called :-
 (1) Haversian canal
 (2) Volkmann's canal
 (3) Central canal
 (4) Inguinal canal
- 171.** Intervertebral discs are composed of :-
 (1) Fibrous cartilage
 (2) Elastic cartilage
 (3) Hyaline cartilage
 (4) Foetal cartilage
- 172.** Ossification is the process by which
 (1) Bone is formed
 (2) Wound healing takes place
 (3) Cartilage is formed
 (4) Skin is formed
- 173.** Ligament connects
 (1) Muscle to muscle
 (2) Muscle to bone
 (3) Integument with muscle
 (4) Bone to bone
- 174.** The cells of cartilage are called
 (1) Chondrocyte
 (2) Odontoblast
 (3) Myocyte
 (4) Osteoblast
- 175.** Tendon connects
 (1) Bone to muscles
 (2) Bone with cartilage
 (3) Nerve to muscle
 (4) Muscle to muscle
- 176.** During an injury nasal septum gets damaged, for recovery which cartilage is responsible?
 (1) Fibrous cartilage
 (2) Calcified cartilage
 (3) Hyaline cartilage
 (4) Elastic cartilage
- 177.** Skeletal tissue found in mammalian pinna is :-
 (1) Fibrous cartilage
 (2) Articular cartilage
 (3) Elastic cartilage
 (4) Calcified cartilage
- 178.** The overstretching of ligaments is termed :-
 (1) Sprain (2) Fatigue
 (3) Tonus (4) Tetany
- 179.** The tissue which forms the basic structure of lymphoid organs, like bone marrow, spleen, etc. is :-
 (1) Areolar tissue (2) Adipose tissue
 (3) Elastic tissue (4) Reticular tissue
- 180.** Histiocyte is a connective tissue cell which takes part in?
 (1) Fibre production (2) matrix production
 (3) Secretion (4) phagocytosis